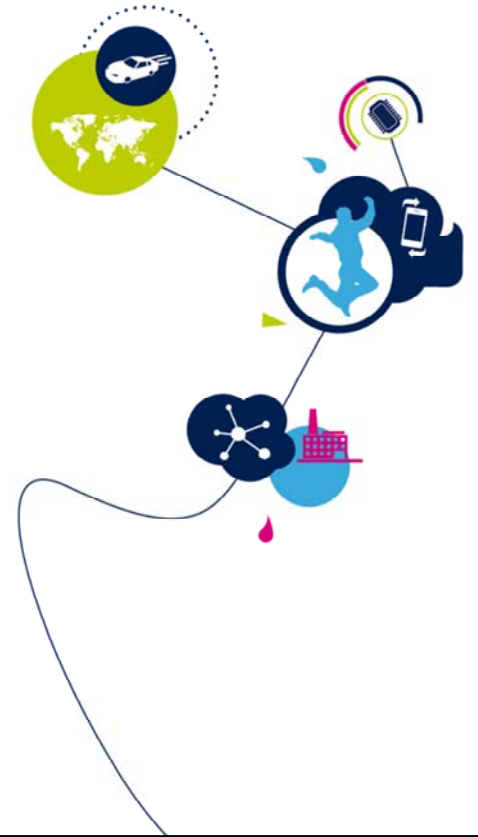


STM32G4 - RCC

Reset and clock controller



Hello, and welcome to this presentation of the STM32G4 reset and clock controller (or RCC).

- The STM32G4 reset and clock controller manages system and peripheral clocks
 - 3 internal oscillators
 - 2 external oscillators (crystal or resonator)
 - One PLL
 - Many peripherals have independent clocks
- The RCC manages the various system and peripheral resets

Application benefits

- High flexibility in choice of clock sources to meet consumption and accuracy requirements
- Many independent peripheral clocks allow for adjusting power consumption without impacting communication baud rates, and to keep some peripherals active in low-power modes
- Safe and flexible reset management



The STM32G4 reset and clock controller manages system and peripheral clocks.

STM32G4 microcontrollers embed 3 internal oscillators, 2 oscillators for an external crystal or resonator, and one phase-locked loop (or PLL).

Many peripherals have their own clock, independent of the system clock.

The RCC also manages the various resets present in the device.

The STM32G4 RCC provides high flexibility in the choice of clock sources, which allows the system designer to meet both power consumption and accuracy requirements.

The numerous independent peripheral clocks allow a designer to adjust the system power consumption without impacting the communication baud rates, and also to keep some peripherals active in low-power

modes.

Finally, the RCC provides safe and flexible reset management.

Safe and flexible reset management without external components

- Manages three types of reset:
 - System reset
 - Power reset
 - Backup domain reset
- Peripherals have individual reset control bits



Safe and flexible reset management without any need for external components reduces application costs.

The RCC manages three types of resets: the system reset, the power reset and the backup domain reset.

The peripherals have individual reset control bits.

- System reset
 - Resets all registers except certain RCC registers, PWR registers, and the Backup domain
- Reset sources
 - Low level on the NRST pin (external reset)
 - WWDG event
 - IWDG event
 - Firewall event
 - A software reset (through NVIC)
 - Low-power-mode security reset
 - Option byte loader reset
 - Brown-out reset
- The Reset Source flag is in the RCC_CSR register



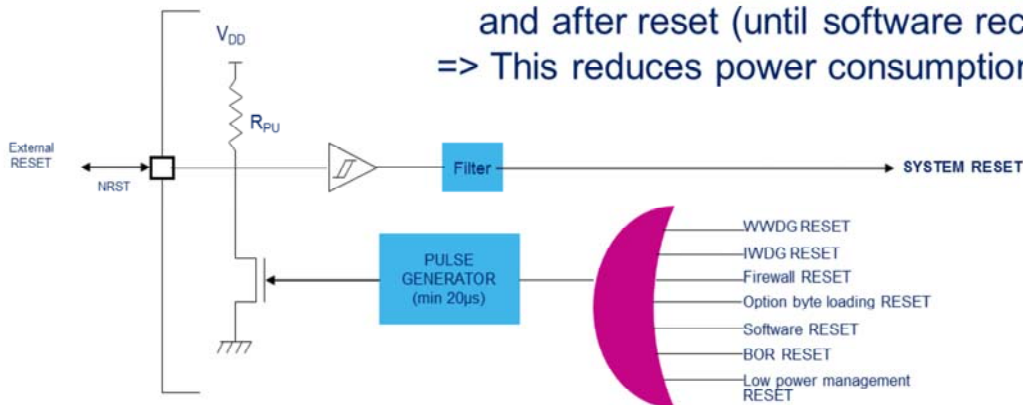
The first type of reset is the System Reset, which resets all the registers except certain registers for the Reset and Clock Controller and Power Controller. It also does not reset the Backup domain.

The system reset sources are the external reset (generated by a low level on the NRST pin), a window watchdog event, an independent watchdog event, a firewall event, a software event through the Nested Vectored Interrupt Controller, a low-power-mode security reset (which is generated when Stop, Standby or Shutdown mode is entered but is prohibited by the option byte configuration), an option byte loader reset, and a brown-out reset.

The reset source flag can be found in the RCC Control and Status register.

No external components are needed due to internal filter and power monitoring
System reset sources can reset external components

- Reset pull-up deactivated during internal reset
 - I/Os in analog mode (no Schmitt trigger) during and after reset (until software reconfiguration).
- => This reduces power consumption under reset



Here is the simplified block diagram of the system reset. All internal reset sources provide a reset signal on the NRST pin, which can be used to reset other components of the application board. In addition, no external reset circuitry is needed due to the internal glitch filter and the safe power monitoring feature which guarantees the reset of the application when VDD is below the selected threshold. The internal pull-up on the NRST pin, which maintains a high level when no reset signal is driven low, is deactivated when an internal reset is driven in order to reduce power consumption under reset. Additionally, all I/O pins are placed in analog mode during and after reset to eliminate power consumption through the Schmitt trigger when the I/Os are floating under reset and before software initialization.

- Power reset
 - Sources
 - Brown-out reset (BOR) => resets all registers except those in the Backup domain
 - Exit from Standby => resets all registers in VCORE domain
 - Registers outside the VCORE domain (RTC, WKUP, IWDG, and Standby/Shutdown mode control) are not impacted.
 - Exit from Shutdown generates a BOR reset.
- Backup domain reset
 - Resets Backup domain RTC registers, Backup registers, and the RCC BDCR register
 - Sources
 - BDRST bit in RCC BDCR register
 - VDD or VBAT power on, if both supplies have previously been powered off



The second type of reset is the power reset. The Brown-out reset (or BOR) resets all registers except those in the Backup domain powered by VBAT which contains the RTC and the external low-speed oscillator. When exiting Standby mode, all registers powered by the regulator are reset. When exiting Shutdown mode, a Brown-out reset is generated.

The third type of reset is the Backup domain reset, which resets the RTC registers, the Backup registers, and the RCC Backup Domain Control Register. This reset occurs when the BDRST bit is set in the RCC Backup Domain control register. It also occurs when VDD and VBAT are powered on if both supplies have previously been powered off.

- NRST Pin new design: PG10-NRST
 - The configuration of the Reset circuitry is done through the option bytes NRST_MODE[1:0] and IRHEN (Internal reset holder option bit)

Mode	Configuration		Behavior
	NRST_MODE	RHEN	
Input/Output (Legacy)	11	0	In case of internal reset, the pulse generator guarantees a minimum reset pulse duration of 20µs to be output on the NRST pin
		1	Ensure that the pin is pulled low until its voltage meets VIL threshold (~0.3VDD)
Input only	01	x	Internal Reset are not propagated outside of the part (PU always ON)
GPIO	10	x	PG10 only, No Reset Pin



Through specific option bits, the NRST pin is configurable for operating as:

- Reset input/output (default at device delivery):

Any valid reset signal on the pin is propagated to device internal logic and all internal reset sources are externally driven through a pulse generator to this pin.

The GPIO functionality (PG10) is not available.

The pulse generator guarantees a minimum reset pulse duration of 20 microseconds for each internal reset source to be output on the NRST pin.

An internal reset holder option can be used, if enabled in the option bytes, to ensure that the pin is pulled low until its voltage meets the VIL threshold.

This function guarantees the detection of internal reset sources by external components when the line faces a significant capacitive load.

In case in an internal reset, the internal pull-up RPU is

deactivated in order to save power consumption through the pull-up resistor.

- Reset input

In this mode, any valid reset signal on the NRST pin is propagated to device internal logic, but resets generated internally by the device are not visible on the pin.

In this configuration, GPIO functionality (PG10) is not available.

- GPIO

In this mode, the pin can be used as PG10 standard GPIO.

The reset function of the pin is not available.

Reset is only possible from device internal reset sources and it is not propagated to the pin.

Choice of clock sources for low-power, accuracy, and performance

- Three internal clock sources
 - High-speed internal 16 MHz RC oscillator (HSI16)
 - High-speed internal 48 MHz RC oscillator (HSI48)
 - Low-speed internal 32 KHz RC oscillator (LSI)
- Two external oscillators
 - High-speed external 4 to 48 MHz oscillator (HSE) with clock security system
 - Low-speed external 32.768 KHz oscillator (LSE) with clock security system
- One PLL, with three independent outputs



The RCC offers a large choice of clock sources, which can be selected depending on the low-power, accuracy, and performance requirements.

STM32G4 devices embed three internal clock sources: a high-speed internal 16 Megahertz RC oscillator (HSI16), a high-speed internal 48 Megahertz oscillator (HSI48), and a low-speed internal 32 kilohertz RC oscillator (LSI).

STM32G4 devices embed two oscillators for use with an external crystal or resonator: a high-speed external 4 to 48 Megahertz oscillator (HSE) with a clock security system and a low-speed external 32.768 kilohertz oscillator (LSE) also with a clock security system.

When enabled, the clock security system can detect failures on external clock sources and an automatic switch to an internal oscillator is performed: HSI16 in case of HSE failure and LSI in case of LSE failure.

If a failure is detected on the HSE clock, a clock failure

event is sent to the break input of the advanced-control timers and to the HRTIM system fault input, and an interrupt is generated to inform the software about the failure.

The PLL present in the STM32G4 has three independent outputs, in order to offer different frequency options to CPU and peripherals.

Clock key features

9

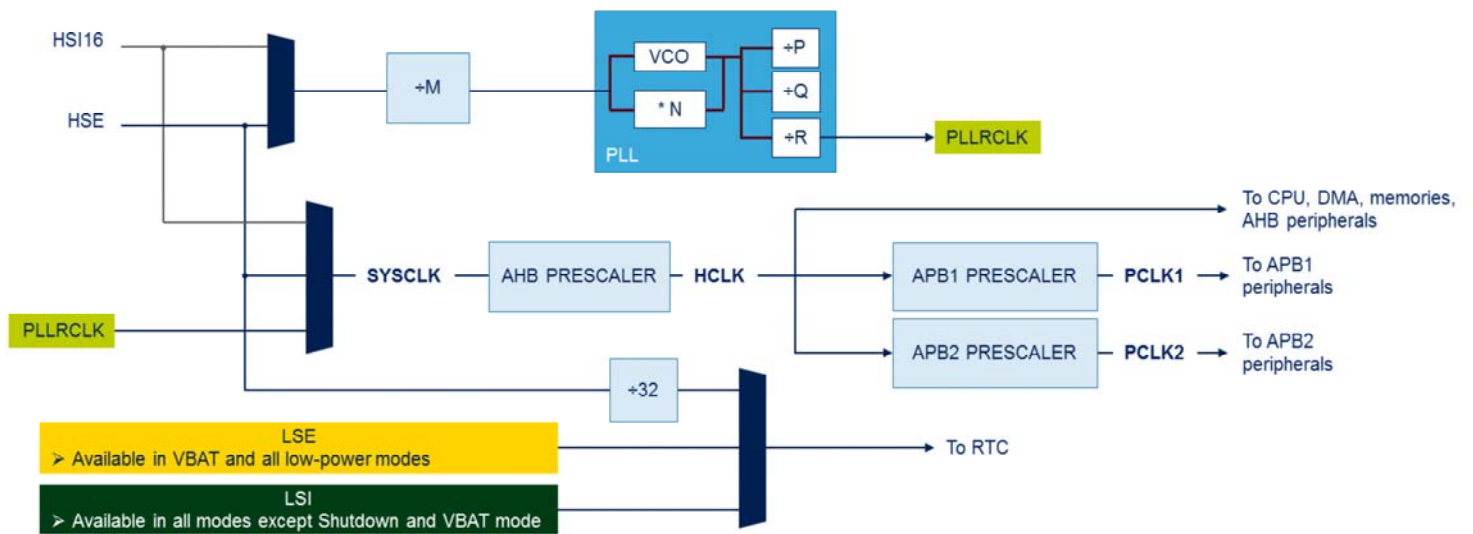
	STM32F3	STM32G4
Internal clock sources	High-speed internal 16 MHz RC oscillator (HSI16)	
	Low-speed internal 40 KHz RC oscillator (LSI)	Low-speed internal 32 KHz RC oscillator (LSI)
	NA	High-speed internal 48 MHz RC oscillator (HSI48)
External clock sources	High-speed external 4 to 32 MHz oscillator (HSE) with clock security system	High-speed external 4 to 48 MHz oscillator (HSE) with clock security system
	Low-speed external 32.768 kHz oscillator (LSE) with clock security system	
PLL	One PLL with a single output	One PLL with three independent outputs



This table highlights the differences between the STM32F3 and the STM32G4 clock sources. Both support the HSI16 and the LSE oscillators. The Low Speed Internal oscillator (LSI) frequency is 40 KHz for the STM32F3, 32 KHz for the STM32G4 microcontroller. The HSI48 is not supported by the STM32F3. The maximum frequency of the HSE is 32 MHz for the STM32F3, 48 MHz for the STM32G4 microcontroller. At last, the STM32G4 PLL has three outputs instead of one in the STM32F3 microcontroller.

Simplified clock tree

10



The system clock can be derived from the high-speed internal 16 MHz RC oscillator (HSI16) or from the high-speed external 4 to 48 MHz oscillator (HSE).

The AHB clock, called HCLK, is derived by dividing the system clock by a programmable prescaler.

The APB clocks, called PCLK1 and PCLK2, are generated by dividing the AHB clock by programmable prescalers.

The RTC clock is generated by the low-speed external 32.768 kHz oscillator (LSE), the low-speed internal 32 kHz RC oscillator (LSI), or the HSE divided by 32.

The LSE can remain enabled in all low-power modes and in VBAT mode.

The LSI can remain enabled in all modes except Shutdown and VBAT modes.

High-Speed Internal (HSI16) clock

11

1% accuracy and fast wakeup time

- HSI16 16 MHz, factory- and user-trimmed
- HSI16 can be selected as
 - Wakeup clock from Stop 0 or Stop 1 modes
 - Backup clock for Clock Security System (CSS)
- Can be automatically started when exiting Stop modes
- I2C, U(S)ART, LPUART can enable the HSI16 during Stop mode to detect their wakeup from Stop sequence.
 - HSI16 remains off during Stop mode except for the peripheral wakeup sequence detection.



The high-speed internal oscillator is a 16 MHz RC oscillator which provides 1% accuracy and fast wakeup times. The HSI16 is trimmed during production testing, and can also be user-trimmed.

The HSI16 can be selected as the clock at wakeup from Stop 0, Stop 1 or Stop 2 modes, and as the backup clock if an HSE failure is detected by the Clock Security System.

The HSI16 can be automatically awoken when exiting Stop mode in order to make it available for peripherals when it is not used as the system clock.

The HSI16 is requested by the I2C and the U(S)ART/LPUART peripherals to support wakeup from Stop 0, Stop 1 or Stop 2 modes. HSI16 is enabled only for the wakeup sequence detection and remains disabled outside of this wakeup sequence.

High-Speed External (HSE) clock

12

Safe crystal system clock

- HSE 4-48MHz
 - External source (Bypass mode) up to 48 MHz
 - External crystal/ceramic resonator (4 - 48 MHz)
- Clock Security System (CSS)
 - Automatic detection of HSE failure with
 - Non-maskable interrupt generation
 - Break input to TIM1/TIM8/TIM20/TIM15/TIM16/TIM17 and `hrtim_sys_flt` => critical applications such as motor control can be put in a safe state.
 - Backup clock is HSI16 => application software does not stop in case of crystal failure.



The high-speed external oscillator provides a safe crystal system clock.

The HSE supports a 4 to 48 MHz external crystal or ceramic resonator, and also an external source in bypass mode.

A clock security system allows an automatic detection of an HSE failure. In this case a Non-Maskable Interrupt is generated, and a break input can be sent to timers in order to put critical applications such as motor control in a safe state. When an HSE failure is detected, the system clock is automatically switched to an internal oscillator, the HSI16, so the application software does not stop in case of crystal failure.

Low-Speed External (LSE) clock

13

32.768 kHz configurable for low-power or high-drive
Available in all power modes and in VBAT mode

- The LSE can be used with external quartz or resonator, or with external clock source in bypass mode
- **Clock Security System on LSE:** Available in all modes except Shutdown and VBAT. Operates under reset.
- The LSE can be used for RTC, U(S)ARTs, LPUART, LPTIMs.

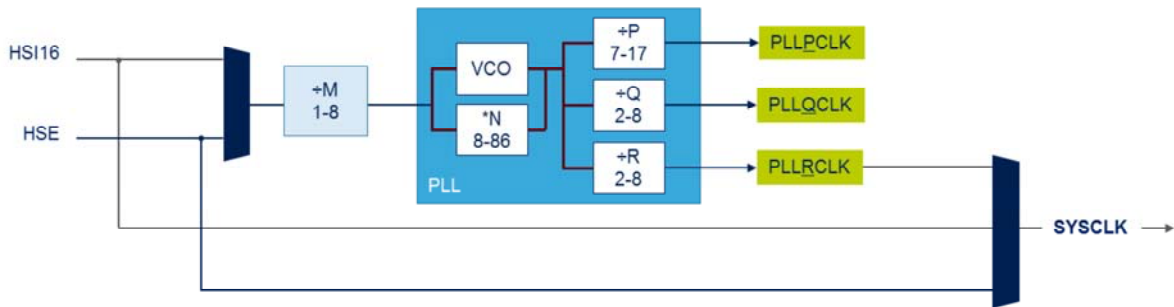
Mode	Maximum critical crystal gm (μ AV)	Consumption (nA)
Ultra-low power	0.5	250
Medium-low driving	0.75	315
Medium-high driving	1.7	500
High driving	2.7	630



The 32.768 kHz low-speed external oscillator can be used with an external quartz or resonator, or with an external clock source in bypass mode. The oscillator driving capability is programmable. Four modes are available, from ultra-low power mode with a consumption of only 250 nanoamps, to high-driving mode.

A clock security system monitors for failure of the LSE oscillator. In case of failure, the application can switch the RTC clock to the LSI. The CSS is functional in all modes except Shutdown and VBAT. It is also functional under reset.

The LSE can be used to clock the RTC, the USARTs or low-power UART peripherals, and the low-power timers.



The input clock of the PLL can be HSI16 or HSE.
 The PLL output called PLLRCLK can provide the system clock.
 The PLL output called PLLQCLK can be selected as the root clock of USB device, RNG, SAI1, QSPI, and FDCAN.
 The PLL output called PLLPCLK can be selected as the root clock of ADCs.

- Selected between HSI16/HSE/PLL
- System clock, AHB, APB1, and APB2 maximum frequency: 170 MHz
- Voltage scaling:

Voltage range		HCLK	HSI16	HSE	PLL
Range 1 boost mode	Main regulator (MR)	∈ [150 MHz, 170 MHz]	16 MHz	48 MHz	170 MHz
Range 1 normal mode		∈ [26 MHz, 150 MHz]	16 MHz	48 MHz	150 MHz
Range 2		≤ 26 MHz	16 MHz	26 MHz	26 MHz
Low-power run/sleep	Low-power regulator	≤ 2 MHz	Allowed	Allowed with divider	Not allowed



The system clock is selected between the HSI16, HSE and PLL R output.

The maximum system clock frequency is 170 MHz.

The APB1 and APB2 bus frequencies are also up to 170 MHz.

The maximum clock source frequency depends on the voltage scaling and power mode.

The system clock is limited to 170 MHz in Range 1 boost mode, 150 MHz in Range 1 normal mode, 26 MHz in Range 2 and 2 MHz in Low-power run/Low-power sleep modes.

- Switching from Low to High speed or from High to Low speed system clock, **it is recommended to use a transition state with a medium speed clock for at least 1 μ s**
- Clock source switching conditions:
 - Switching from HSE or HSI to PLL with AHB frequency (HCLK) higher than 80 MHz
 - Switching from PLL with HCLK higher than 80 MHz to HSE or HSI
- Transition state to increase the frequency:
 - Set the AHB prescaler HPRE[3:0] bits to divide System frequency by 2
 - Switch system clock to PLL
 - Reconfigure AHB prescaler bits to needed HCLK frequency



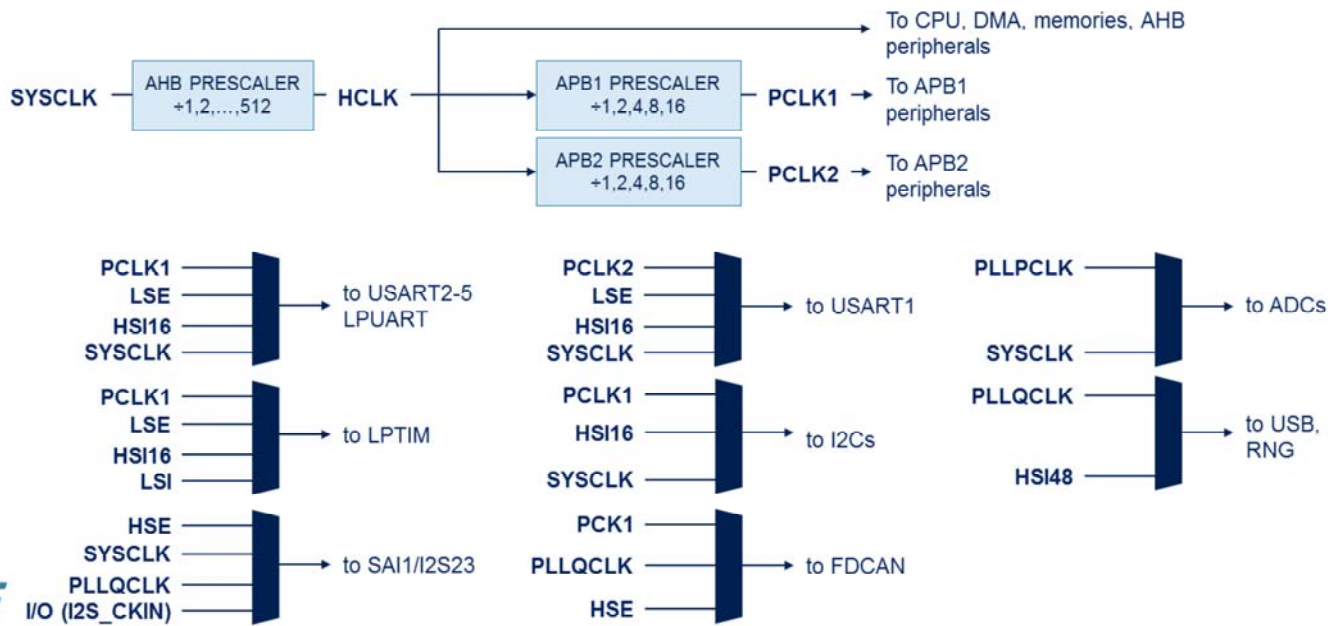
In STM32G4 devices, it is recommended to use a transition state while switching from Low to High speed or from High to Low speed system clock.

This slide presents the recommended sequence for the transition state.

To increase the frequency, the AHB clock frequency has to be divided by two prior to switching the system clock to PLL.

After a one microsecond delay, the prescaler providing the AHB clock can be set to the targeted frequency.

This is needed when switching from HSE or HSI to PLL and the system frequency becomes higher than 80 MHz.

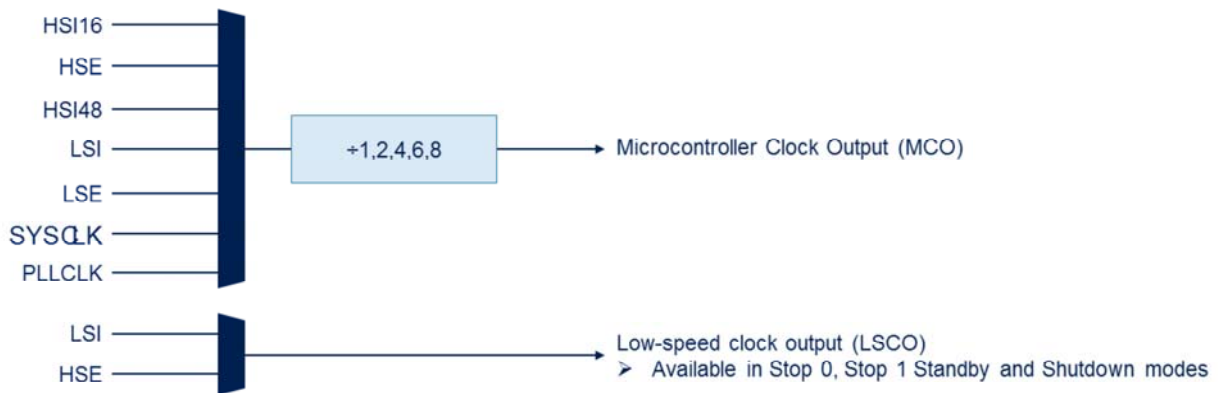


The AHB prescaler divides the system clock to obtain the AHB clock (HCLK).

APB clocks, called PCLK1 and PCLK2 are obtained by applying a programmable prescaler ratio to HCLK.

This slide also describes the multiplexers in charge of selecting the clock of various peripherals.

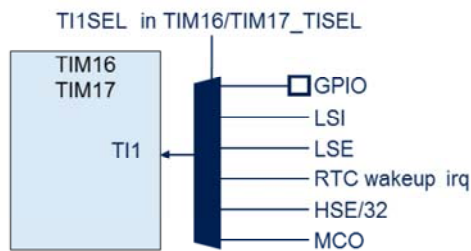
The ADC clock is derived from the system clock, or from the PLL "P" output. This PLLPCLK has no other usage.



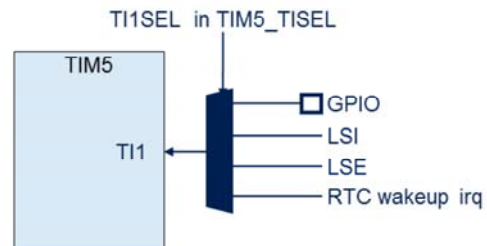
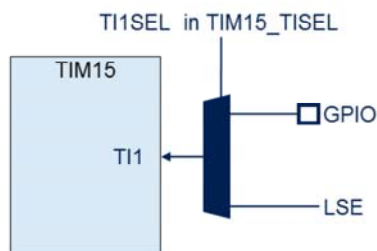
The various clocks can be output on an I/O pad.
The Microcontroller Clock Output feature enables the external output of one of these seven clocks: HSI16, HSI48, HSE, LSI, LSE, SYSCCLK, and PLLCLK.
The low-speed clock output feature enables the external output of the LSI or LSE clock.
The low-speed clock output is available in Stop 0, Stop 1, Standby and Shutdown modes.

Internal/external clock measurement with Timers

19



- Each device is calibrated at the ST factory for 1 % accuracy at the ambient temperature of 25°C



It is possible to indirectly measure the frequency of all on-board clock sources by means of the TIM5, TIM15, TIM16 or TIM17 channel 1 input capture.

These measurements can be used to calibrate the LSI and HSI16 frequencies.

Regarding HSI16, RC oscillator frequencies can vary from one chip to another due to manufacturing process variations, this is why each device is calibrated at the ST factory for 1 % accuracy at the ambient temperature of 25°C

After reset, the HSI16 factory calibration value is automatically loaded.

If the application is subject to voltage or temperature variations, software can perform a clock trimming.

Dynamic consumption optimization in (LP)Run and (LP)Sleep modes

- Peripheral clock enable registers
 - Peripherals clocks disabled by default (except Flash memory)
 - Registers read and write access not supported when the clock is disabled
 - Caution: SRAM1/SRAM2/CCM SRAM internal memories do not have enable bit (always enabled in Run/LPRun modes)
- Peripheral clock enable registers in Sleep and Stop modes
 - Enables or disables the peripheral clocks in Sleep, LPSleep, Stop 0/1 modes
 - No effect if the corresponding peripheral clock enable bit is cleared
 - Controls both bus and kernel clocks
 - Affects Sleep and Stop modes (for peripheral with independent clock active in Stop mode)
 - Caution: SRAM1/SRAM2/CCM SRAM clocks are enabled by default in Sleep/LPSleep modes



The dynamic power consumption can be optimized by using peripheral clock gating.

Each peripheral clock can be gated ON or OFF in Run and Low-power run mode, except SRAM1, SRAM2 and CCM SRAM internal memories which are always clocked in Run and Low-power run modes.

By default, the peripheral's clock is disabled, except the Flash memory clock which is enabled.

When a peripheral's clock is disabled, the peripheral's registers cannot be read or written.

Other registers allow the configuration of the peripheral's clock during the Sleep and Low-power sleep modes.

This also affects Stop 0 and Stop 1 modes for peripherals with an independent clock active in Stop modes.

These control bits have no effect if the corresponding peripheral clock enable bit is cleared.

By default the SRAM1, SRAM2 and CCM SRAM clocks are enabled in Sleep and Low-power sleep modes. If they are not needed, the SRAM clock enable bits should be disabled to reduce power consumption.

Interrupt event	Description
LSE clock security system	Set when a failure is detected in the LSE oscillator
HSE clock security system	Set when a failure is detected in the HSE oscillator
PLL ready interrupt flag	Clock ready caused by PLL lock
HSE ready	Clock ready caused by the HSE oscillator
HSI16 ready	Clock ready caused by the HSI16 oscillator
HSI48 ready	Clock ready caused by the HSI48 oscillator
LSE ready	Clock ready caused by the LSE oscillator
LSI ready	Clock ready caused by the LSI oscillator



This slide lists the RCC interrupts. The LSE and HSE clock security systems, the PLL ready, and all five oscillator ready signals can generate an interrupt.

- Refer to these trainings linked to this peripheral, if any
 - STM32G4 Power control (PWR)
 - STM32G4 Interrupts (NVIC-EXTI)



In addition to this training, you may find the Power Control and Interrupt Controller trainings useful.

- For more details, please refer to following sources
 - STM32G4xx reference manual RM0440
 - AN2867 Oscillator design guide for STM8S, STM8A and STM32 microcontrollers
 - AN4736 How to calibrate STM32L4 internal RC oscillators



For more details, please refer to application note AN2867, an oscillator design guide for STM8S, STM8A and STM32 microcontrollers and application note AN4736 which explains how to calibrate STM32L4 internal RC oscillators.